ASSIGNMENT

DIPLOMA IN INFORMATION TECHNOLOGY



Student Full Name	:	Thanthrige Dinara Lithmini Perera
Name with Initial	:	T.D.L Perera
Register Number	:	-
Student ID Number	:	PSBU/SITC/DIP/ICT/SL0016832
NIC Number	:	200251000269
Contact Number	:	+94 70 397 5381
WhatsApp Number	:	+94 70 397 5381

SITC CAMPUS

Table of Contents

Development of a Basic Calculator Using Python 3
1. Introduction 3
2. Objectives 3
3. Software used 3
4. Methodology 4
   4.1 Functional requirements 4
   4..2 Mathematical Logic and Decision Making 4
5. Implementation 4
   5.1 Error and Exception Handling 4
   6. Screenshots of Program Execution 4
   7. Source Code 6
   8. Output Result 7
   9. Conclusions 8

Figure 1: Addition Test Execution 4
Figure 2:Multiplication Test Execution 5
Figure 3:Division by Zero Guard Test 5
Figure 4:Source Code 6
Figure 5: Addition output result 7
Figure 6: Subtraction output result 7
Figure 7: Multiplication output result 7
Figure 8: Division output result 7

Development of a Basic Calculator Using Python

1. Introduction

Python is currently one of the most popular computer languages in the world due to its ease of reading and ability to perform a variety of tasks. This is mostly used for software development, data science and automation. In this project, the computer language used is Python. Based on it, a basic calculator application that runs through the terminal has been created.

To get user inputs, perform mathematical operations, and display accurate results, a calculator application requires a proper logic system. Python provides powerful conditional statements and error handling structures. Therefore, it is a great language to build a safe and user-friendly calculator.

2. Objectives

The main objectives of this assignment are:

- To design and run a working calculator program on the computer terminal.
- To get two numbers (no1 and no2) and a math sign (+, -, *, /) from the user.
- To use simple rules (if-elif-else) to do the correct math calculations.
- To stop the programming from crashing when someone tries to divide a number by zero
- To create a clean and neat assignment report that follows SITC Campus guidelines.

3. Software used

Tool name	Version	Purpose
Python Interpreter	v3.12.x	Core programming language runtime and script engine.
JetBrains PyCharm	Community Edition	Integrated Development Environment (IDE) used for writing and executing code.
Microsoft Word	2026 / Office 365	Technical word processor used to write and format this assignment report.

4. Methodology

4.1 Functional requirements

The program starts by asking the user to type two numbers. It saves these numbers as floats so the calculator can work with decimal numbers too. Then, it asks the user to choose a math sign ($+, -, *, /$) to know what to calculate.

4..2 Mathematical Logic and Decision Making

After getting the numbers and the sign, the program checks the sign using if-elif-else rules. If the sign is +, it adds the numbers. If it is -, it subtracts them. If it is *, it multiplies them. Finally, the program calculates the answer and shows it on the screen.

5. Implementation

5.1 Error and Exception Handling

In mathematics, we cannot divide a number by zero (0). In computer programs, if you try to divide by zero, the program will crash instantly.To follow the assignment rules, we added a special check to stop this mistake. Before dividing, the program checks: if no2 == 0 If the second number is zero, the program shows an error message instead of crashing. Also, we used a try-except rule. This safely warns the user if they type letters instead of numbers.

6. Screenshots of Program Execution

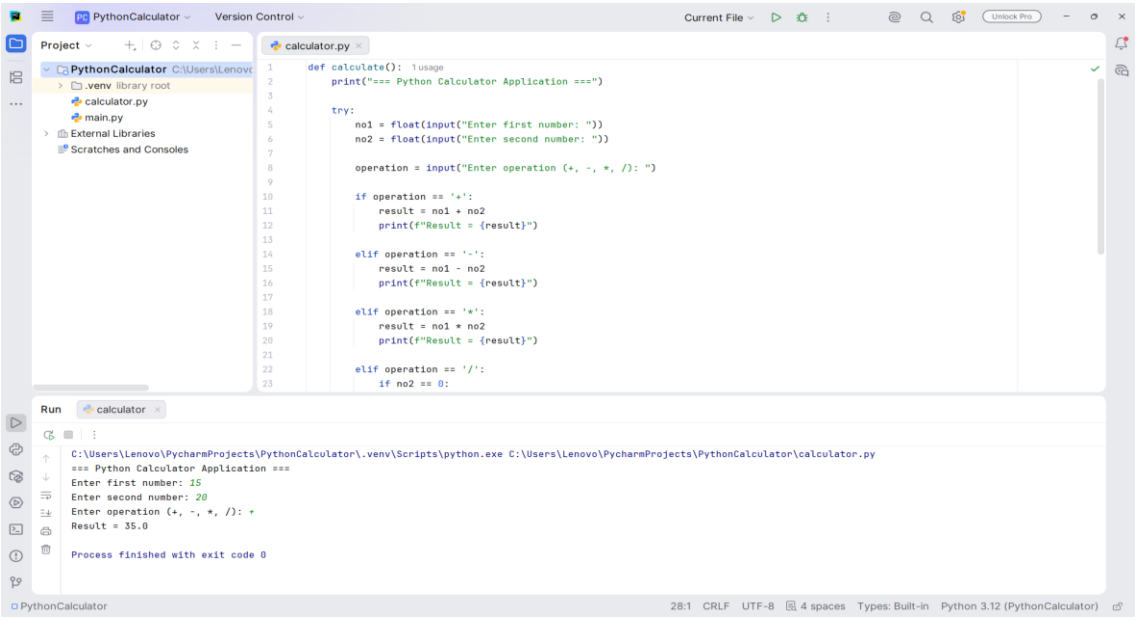


Figure 1: Addition Test Execution

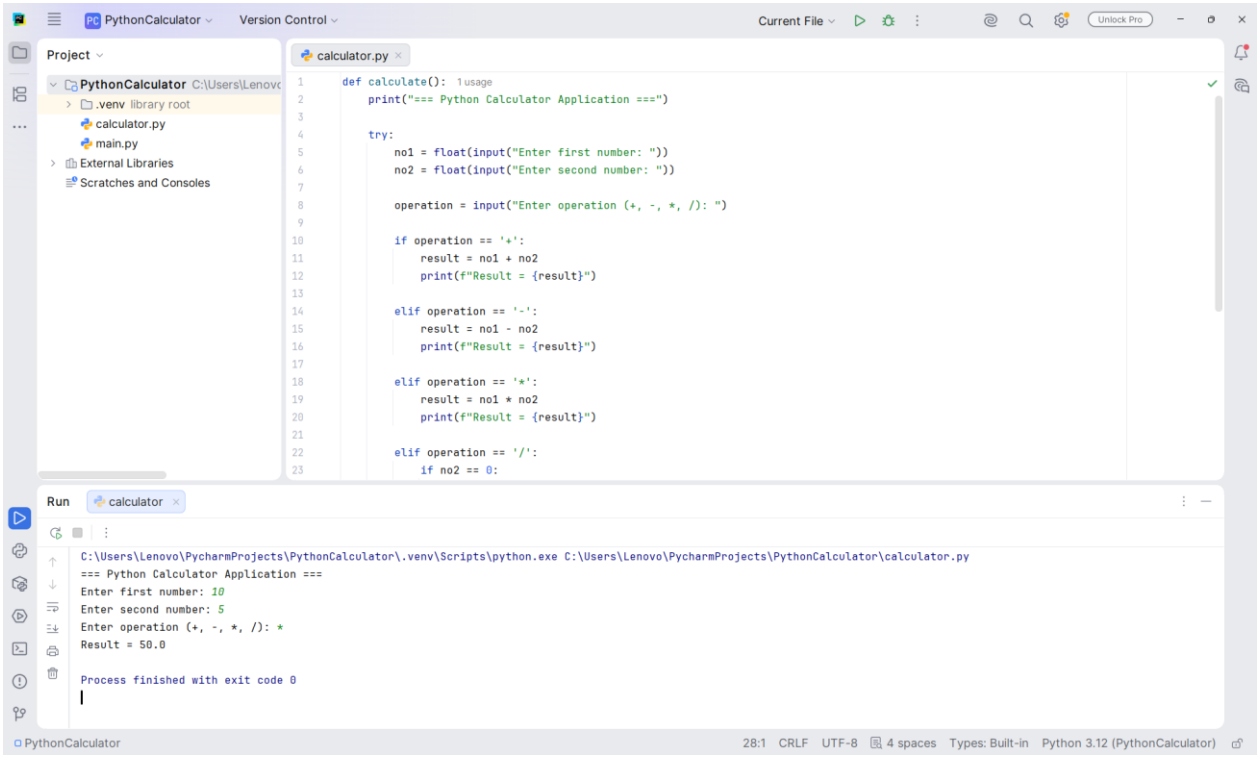


Figure 2:Multiplication Test Execution

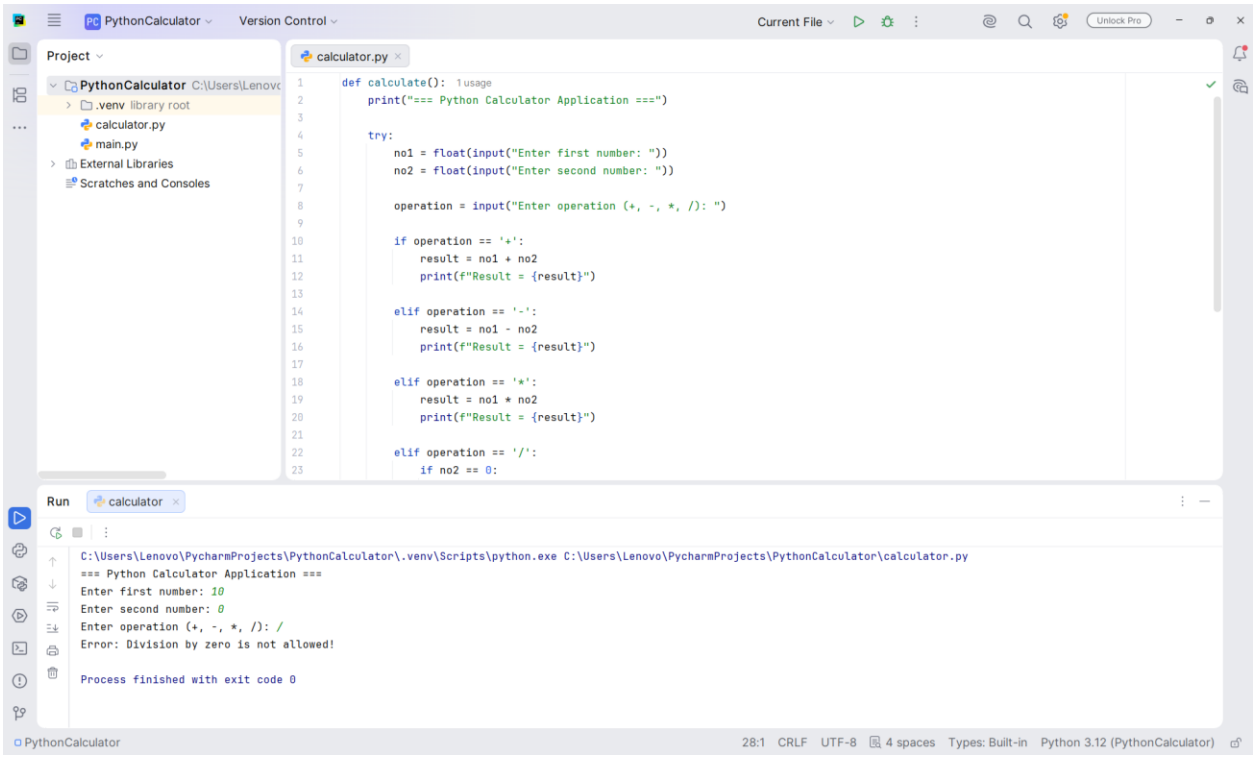


Figure 3:Division by Zero Guard Test

7. Source Code

```
def calculate():  1 usage
    print("=== Python Calculator Application ===")

    try:
        no1 = float(input("Enter first number: "))
        no2 = float(input("Enter second number: "))

        operation = input("Enter operation (+, -, *, /): ")

        if operation == '+':
            result = no1 + no2
            print(f"Result = {result}")

        elif operation == '-':
            result = no1 - no2
            print(f"Result = {result}")

        elif operation == '*':
            result = no1 * no2
            print(f"Result = {result}")

        elif operation == '/':
            if no2 == 0:
                print("Error: Division by zero is not allowed!")
            else:
                result = no1 / no2
                print(f"Result = {result}")

        else:
            print("Invalid operation! Please enter +, -, *, or /.")

    except ValueError:
        print("Invalid input! Please enter valid numbers.")


calculate()
```

Figure 4:Source Code

8. Output Result



Figure 5: Addition output result



Figure 6: Subtraction output result



Figure 7: Multiplication output result



Figure 8: Division output result

9. Conclusions

I have fully completed Task 1 of the Diploma in Information Technology assignment. By building this calculator program, I learned how to get user inputs, use if-elif-else rules, and handle errors (like division by zero). The program works perfectly in PyCharm and does not crash when there is a mistake.